Name: Dharmit Kunal Shah **PRN: 25070122073** **CSE A-2**

Subject: Programming with Java

ASSIGNMENT No : 11

TITLE : Write a Java program to demonstrate the implementation of interfaces.

Problem Statement

Write a Java program to demonstrate the implementation of interfaces.

Learning Objectives

To implement interfaces for achieving abstraction and multiple inheritance of type.

Theory

An interface specifies a set of abstract methods that implementing classes must define. It provides complete abstraction and supports multiple inheritance of type. Interfaces improve modularity, loose coupling, and software flexibility. They allow different classes to implement common behavior while maintaining independent implementations. Interfaces are widely used in enterprise application development.

```java
interface Animal {
    void sound();
}

class Dog implements Animal {
    public void sound() {
        System.out.println("Dog barks.");
    }
}

public class Main {
    public static void main (String[] args) {
        Animal a = new Dog();
        a.sound();
    }
}
```

```
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp11> java Main.java
Dog barks.
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp11>
```

Exercise

1. Create an interface Printable and implement it in Student and Employee classes.

```java
interface Printable {
    void print();
}

class Student implements Printable {
    String name;
    int rollNo;

    Student(String name, int rollNo) {
        this.name = name;
        this.rollNo = rollNo;
    }

    public void print() {
        System.out.println("Student Name : " + name);
        System.out.println("Roll No      : " + rollNo);
    }
}

class Employee implements Printable {
    String name;
    int empId;

    Employee(String name, int empId) {
        this.name = name;
        this.empId = empId;
    }

    public void print() {
        System.out.println("Employee Name : " + name);
        System.out.println("Employee ID   : " + empId);
    }
}

public class Ex1 {
    Run | Debug
    public static void main(String[] args) {
        Printable s = new Student(name: "Dharmit", rollNo: 158);
        Printable e = new Employee(name: "Ishaan", empId: 501);

        s.print();
        System.out.println();
        e.print();
    }
}
```

```
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp11\Exercise> java Ex1.java
Student Name : Dharmit
Roll No      : 158

Employee Name : Ishaan
Employee ID   : 501
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp11\Exercise>
```

2. Create an interface Switchable with a method turnOn(). Implement the interface in Light and Fan classes to display the device status.

```java
interface Switchable {
    void turnOn();
}

class Light implements Switchable {
    public void turnOn() {
        System.out.println(x: "Light is ON");
    }
}

class Fan implements Switchable {
    public void turnOn() {
        System.out.println(x: "Fan is ON");
    }
}

public class Ex2 {
    Run | Debug
    public static void main(String[] args) {
        Switchable light = new Light();
        Switchable fan = new Fan();

        light.turnOn();
        fan.turnOn();
    }
}
```

```
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp11\Exercise> java Ex2.java
Light is ON
Fan is ON
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp11\Exercise>
```